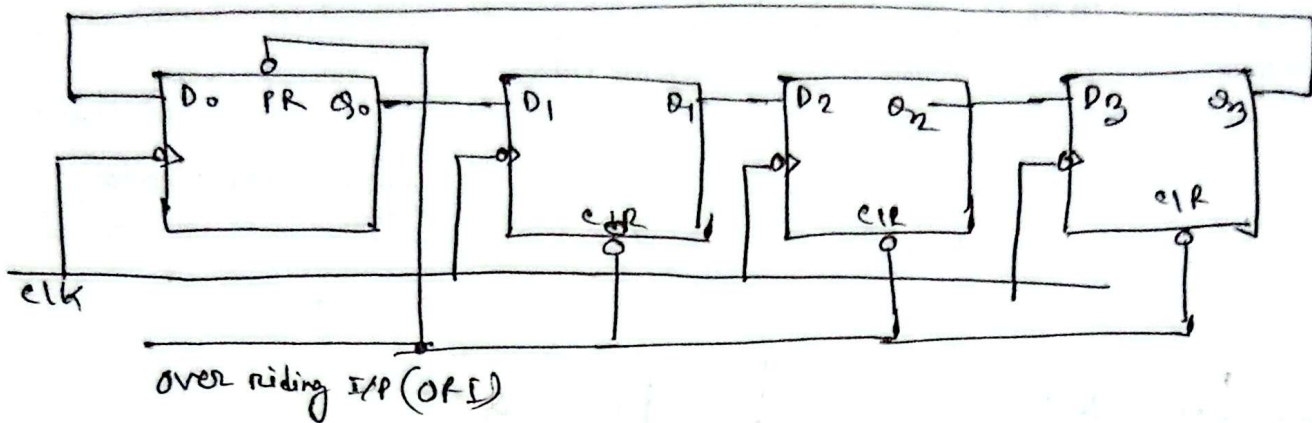


Ring Counter

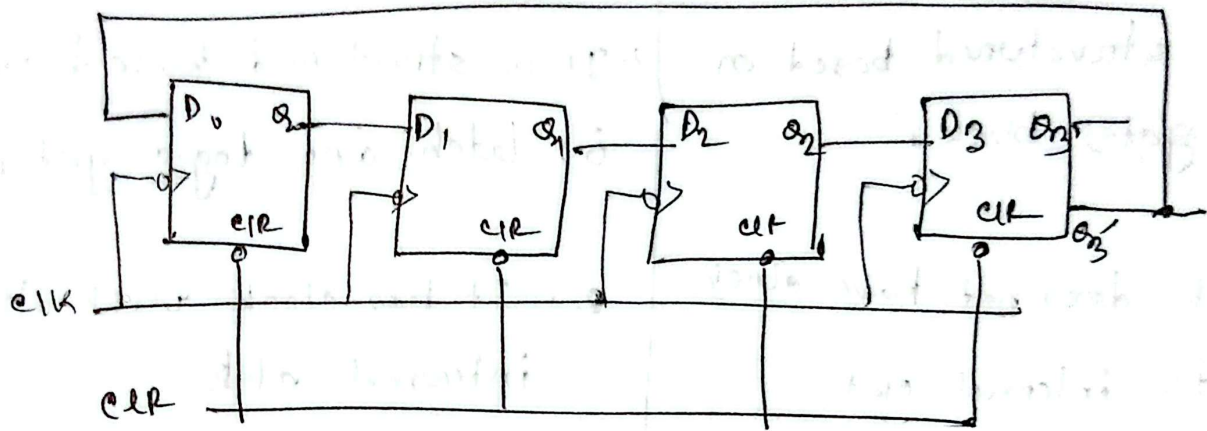
→ It is an application of shift register



when giving active low signal as an over riding i/p to the PR
no matter the i/p Q_0 will be 1, and clr makes $Q_1 = 0$
 $Q_2 = 0$, $Q_3 = 0$

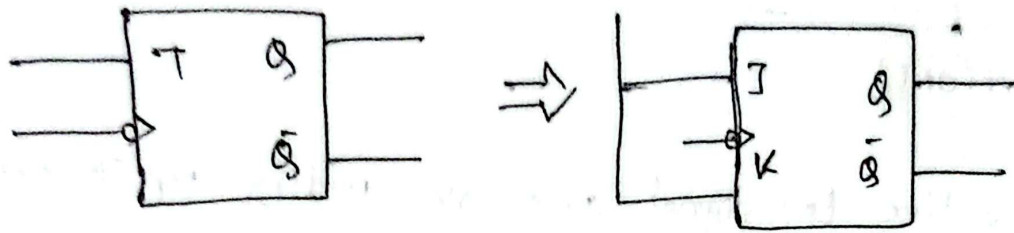
OR.I	clk	Q_0	Q_1	Q_2	Q_3
	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

Johnson's Counter:-



clr	clk	Q ₀	Q ₁	Q ₂	Q ₃
	X	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	1
1	↓	0	0	0	0

$$Q_3' = 1 \rightarrow D = 1 \rightarrow Q_0$$



Modulo Counter :-

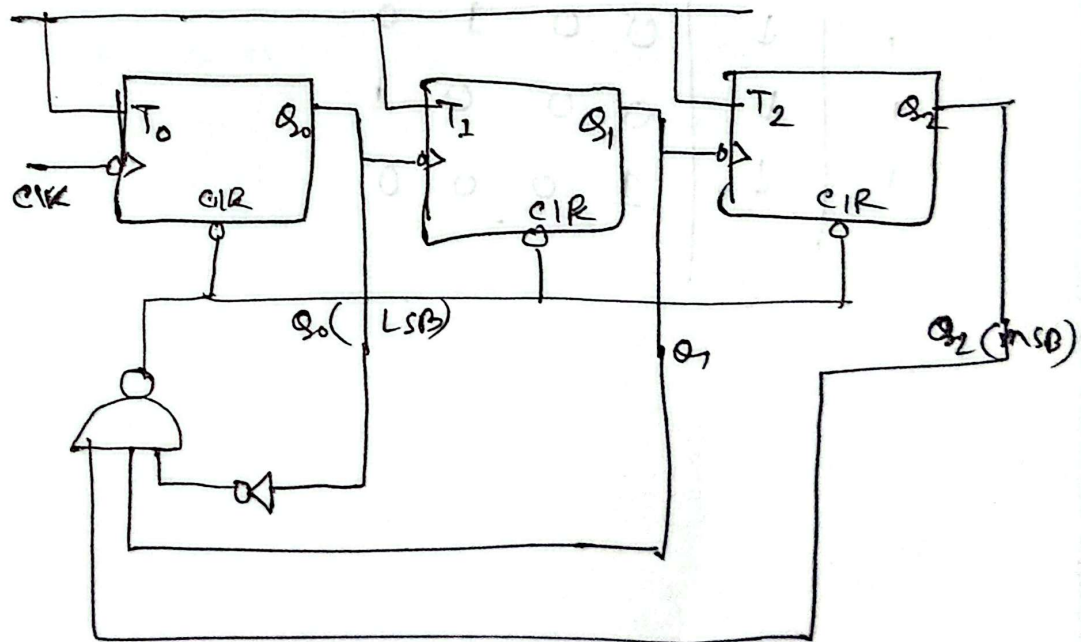
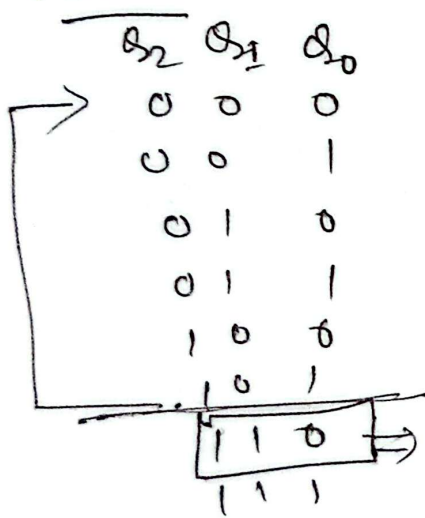
mod 6 counter (0 to 5 counts)

bits 3 ($2^3 = 8$)

mod 12 " (0 to 11 counts)

bits 4 ($2^4 = 16$)

Mod 6:



Counter

- UP counter $[0, 1, 2, \dots, N]$
- Down Counter $[N, N-1, \dots, 1, 0]$

classification

- 1) Synchronous Counter (memory elements have same clock)
 - 2) Asynchronous Counter (memory elements have different clock pulse)
- For n bits counter, max count $= 2^n$

Ripple Counter (Asynchronous Counter)

